

Hahn–Tate Uniformization at Arbitrary Valuation Rank

Exact summability domains, surcomplex elliptic curves,
and torsion ramification

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Abstract

Let $K = \mathbb{C}((t^\Gamma))$ be a Hahn field, with no restriction on the rank or divisibility of the set-sized ordered abelian group Γ . For $q \in K$ of positive valuation α , we identify the exact domain on which the classical bilateral theta and Tate coordinate families are strongly Hahn-summable: it is the multiplicative subgroup U_q whose valuations belong to the principal convex subgroup generated by α . Outside this domain, leading exponents themselves contain a strictly decreasing sequence. Nevertheless, the resulting map gives an isomorphism $U_q/q^\mathbb{Z} \simeq E_q(K)$ onto *all* rational points of the Tate elliptic curve. The proof combines a rank-one coarsening inside the principal convex subgroup with unrestricted formal evaluation at infinitesimals for a second, coarser valuation. It therefore does not mistake Hahn summation for convergence at every surreal scale.

We also prove compatibility with exponent embeddings, an exact reduction sequence, a valuation-circle quotient, support-preserving recovery of q from j , and the formula $K(E_q[n]) = K_{\Gamma+\mathbb{Z}\alpha/n}$ for the full torsion field. Every elliptic curve over $\mathbf{No}[i]$ with negative valuation of its j -invariant is covered, using set-sized normal-form workspaces. Explicit rank-two and genuinely surcomplex examples exhibit both the forbidden parameter scales and the arbitrarily finer infinitesimals that the parametrization still captures.

Status and scope. This manuscript takes the *new-theorems* alternative, not a claimed solution of a named published open problem. The candidate original contribution is the exact-domain, arbitrary-rank uniformization package and its support-theoretic proof. Tate’s rank-one theorem, theta identities, and the classical torsion construction are credited as inputs. A targeted literature and repository audit found no matching formulation, but does not certify priority. The proofs have not been independently refereed or formalized in Lean. Included exact computations check finite identities, not the infinite arguments.

Keywords. Surcomplex numbers; Hahn series; Tate curve; convex valuation subgroup; strong summability; formal groups; ramification.

Contents

1	The problem and the principal results	3
1.1	Why the result is not an entire-function substitution	4
1.2	Relation to the repository and to the literature	5
2	Hahn summation and the relevant order structure	5
2.1	Strong summability	5
2.2	Normalization and two convex subgroups	6
3	The exact domain of the bilateral theta series	7
4	Tate coordinates, support certificates, and precision	8
4.1	A form with positive tails	8
4.2	The elliptic curve and its invariant differential	9
5	Coarse infinitesimals admit unrestricted formal evaluation	10
5.1	The coarse residue field	10
5.2	The rank-one field at the q -scale	11
6	Formal germs and the passage from rank one to arbitrary rank	12
6.1	The classical input, stated with its hypotheses	12
6.2	The formal neighborhood of the identity	12
6.3	Taylor prolongation through coarse infinitesimals	13
7	Proof of arbitrary-rank uniformization	14
7.1	The curve equation and the group law	14
7.2	Good reduction for the coarse valuation	14
7.3	Surjectivity	15
7.4	The kernel	15
7.5	The exact reduction diagram	15
8	Scalar extension and the valuation-circle quotient	16
8.1	Strong functoriality	16
8.2	A canonical circularly ordered value quotient	17
9	Recovering the Tate parameter without adding scales	17
9.1	The inverse modular series	17
9.2	Which elliptic curves this covers	18
10	Torsion fields and exact exponent ramification	18

10.1	Roots of a Hahn parameter	19
10.2	A primitive torsion point	19
11	Transport to surreal and surcomplex numbers	20
11.1	Set-sized normal-form workspaces	20
11.2	A forbidden monomial and an admitted fine displacement	21
11.3	Why fine-topological convergence is not being used	21
12	Exact computation, proof dependencies, and implementation	21
12.1	Finite checks included with this manuscript	21
12.2	What an exact symbolic implementation must know	22
12.3	A dependency-aware formalization plan	22
13	Limitations and further mathematical questions	23
14	Conclusion	24
A	Source and verification audit	24
A.1	What is attributed, and what is proved here	24
A.2	Repository provenance	24
A.3	Literature-search scope	25
A.4	Reproduction	25

1 The problem and the principal results

The classical multiplicative uniformization of an elliptic curve uses a parameter $u \in k^\times$ modulo powers of a small element q . Over a complete real-valued non-Archimedean field, Tate's theorem supplies the isomorphism $k^\times/q^\mathbb{Z} \simeq E_q(k)$ [1, 2]. Replacing k by a Hahn field of arbitrary rank is not a harmless change of coefficients. A positive value $\alpha = v(q)$ need not dominate all other values by its integer multiples. The same bilateral series can then have non-well-ordered support.

There are two different questions. Which parameters admit the literal Hahn formulas? And, after excluding the inadmissible parameters, do those formulas still reach every elliptic-curve point? The second question does not follow from a support calculation. It is the central issue addressed here.

Throughout, Γ is a set-sized ordered abelian group, written additively, and

$$K = K_\Gamma := \mathbb{C}((t^\Gamma)).$$

The coefficient field $\mathbb{C}$ has the trivial valuation. Every integer, degree, and index $n \in \mathbb{Z}$ in a series is an *ordinary finite* integer, not a surreal integer. We use $v(t^\gamma) = \gamma$ and $v(0) = +\infty$.

Definition 1.1 (The admissible group). For $q \in K^\times$ with $\alpha = v(q) > 0$, put

$$H_\alpha = \{\gamma \in \Gamma : -N\alpha \leq \gamma \leq N\alpha \text{ for some } N \in \mathbb{N}\}, \quad (1)$$

$$U_q = v^{-1}(H_\alpha) \subseteq K^\times. \quad (2)$$

Thus H_α is the smallest convex subgroup of Γ containing α . We call U_q the *Hahn-admissible multiplicative domain*.

Define the standard Tate coefficients

$$s_k(q) = \sum_{n \geq 1} \frac{n^k q^n}{1 - q^n}, \quad (3)$$

$$a_4(q) = -5s_3(q), \quad a_6(q) = -\frac{5s_3(q) + 7s_5(q)}{12}, \quad (4)$$

$$E_q : y^2 + xy = x^3 + a_4(q)x + a_6(q). \quad (5)$$

The point at infinity is denoted O . These definitions are legitimate strong Hahn sums; the discriminant is nonzero by theorem 4.3.

Theorem 1.2 (Exact-domain Hahn–Tate uniformization). *Let $q \in K_\Gamma$ satisfy $v(q) = \alpha > 0$. The bilateral family*

$$\Theta_q(u) = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(n-1)/2} u^n \quad (6)$$

is strongly Hahn-summable if and only if $u \in U_q$. For $u \notin q^\mathbb{Z}$, each of the two bilateral families defining

$$X_q(u) = \sum_{n \in \mathbb{Z}} \frac{q^n u}{(1 - q^n u)^2} - 2s_1(q), \quad (7)$$

$$Y_q(u) = \sum_{n \in \mathbb{Z}} \frac{(q^n u)^2}{(1 - q^n u)^3} + s_1(q) \quad (8)$$

is strongly Hahn-summable if and only if $u \in U_q$.

The map

$$\Phi_q(u) = \begin{cases} (X_q(u), Y_q(u)), & u \notin q^{\mathbb{Z}}, \\ O, & u \in q^{\mathbb{Z}}, \end{cases} \quad (9)$$

is a surjective group homomorphism $U_q \rightarrow E_q(K)$ with kernel $q^{\mathbb{Z}}$. In particular,

$$\boxed{U_q/q^{\mathbb{Z}} \simeq E_q(K_{\Gamma})}. \quad (10)$$

No divisibility, countability, finite-rank, or real-valuedness assumption on Γ is needed.

The summability assertions are proved in sections 3 and 4; the group and surjectivity assertions are proved in section 7. The qualification “the bilateral families” matters: this is an exact maximality theorem for specified expressions, not a claim that no different function can be assigned outside U_q .

Theorem 1.3 (Exact arithmetic and scalar extension). *For an integer $n \geq 2$, let $\Lambda = \Gamma + \mathbb{Z}\alpha/n \subseteq \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$. Inside $K_{\Gamma \otimes \mathbb{Q}}$ one has*

$$K_{\Gamma}(E_q[n]) = K_{\Gamma}(q^{1/n}) = K_{\Lambda}, \quad [K_{\Lambda} : K_{\Gamma}] = \text{ord}(\alpha + n\Gamma) \leq n. \quad (11)$$

A single point $\Phi_q(q^{1/n})$ generates this field through its coordinates. The extension is cyclic and totally ramified.

An ordered embedding of exponent groups preserves the Hahn formulas and Φ_q . In a larger Hahn field the exact parameter domain is again the inverse image of the principal convex subgroup generated by the image of α ; it need not be the whole multiplicative group.

The torsion description is classical in its Kummer origin. The point of theorem 1.3 is the explicit arbitrary-group ramification formula and its compatibility with the domain theorem, not a claim to have discovered Tate torsion.

Corollary 1.4 (Surcomplex elliptic curves). *Fix the Conway normal-form valuation on $\mathbf{No}[i]$, with $v(\omega^{-\gamma}) = \gamma$. For every nonzero q with $\alpha = v(q) > 0$,*

$$\{u \in \mathbf{No}[i]^{\times} : |v(u)| \leq N\alpha \text{ for some } N \in \mathbb{N}\}/q^{\mathbb{Z}} \simeq E_q(\mathbf{No}[i]). \quad (12)$$

Every elliptic curve E over $\mathbf{No}[i]$ with $v(j(E)) < 0$ is isomorphic to an E_q of this kind. The parameter q is uniquely determined by $j(E)$. These are class statements whose every instance and proof is realized in a set-sized Hahn workspace.

1.1 Why the result is not an entire-function substitution

If $H_{\alpha} = \Gamma$, the parameter domain is all of $K^{\times}$. For a nonzero real-valued group this always happens. If $\Gamma = \mathbb{R} \oplus_{\text{lex}} \mathbb{R}$ and $\alpha = (0, 1)$, however, $H_{\alpha} = \{0\} \oplus \mathbb{R}$. Elements of valuation $(1, 0)$ or $(-1, 0)$ are excluded. Nevertheless, all K -points of E_q are represented. In particular the restriction does not mean that only points whose coordinates have q -bounded valuations are reached.

The accepted parameter $u = 1 + h$, where $v(h)$ dominates every $N\alpha$, already produces points with

$$X_q(1 + h) \sim h^{-2}, \quad Y_q(1 + h) \sim -h^{-3}.$$

Their coordinates can be vastly larger than every integral power of q^{-1} . Smallness of u and smallness of $u - 1$ are different questions.

1.2 Relation to the repository and to the literature

The repository audit was targeted: its document map, the root description, the spectral report, and the rank-one analytic report were inspected, and code search was used for elliptic/Tate terminology. The pinned revision is a3124af79f66b8b9c196d76b4cbc5ac3938907c4 [15]. The closest existing report works over the fixed field $\mathbb{C}((t^{\mathbb{R}}))$ and develops Tate *algebras*, disks, annuli, and a partial-theta example [16]. A Tate algebra is not a Tate elliptic curve. The present paper neither reclaims that analytic infrastructure nor repeats the partial-theta zero calculation.

That report is emphatic that its base is the fixed rank-one field $\mathbb{C}((t^{\mathbb{R}}))$, and it does not treat higher rank as a cosmetic variant. Its `prop:rankobstruction` proves that the ordered group $\mathbb{Q} + \mathbb{Q}\omega \subset \mathbf{No}$ admits no order-preserving injective additive map to $\mathbb{R}$: since $n \cdot 1 < \omega$ for every positive integer n , an Archimedean target cannot receive both scales faithfully. A higher-rank valuation therefore cannot be silently scalarized; as that report puts it, a noninjective order-preserving homomorphism discards an entire layer of scale, and in its example it must kill 1. The present paper is the arbitrary-rank work that [16] explicitly declines to attempt, carried out on a different object. We take the obstruction as binding rather than as an obstacle to be argued away: the real-valued coarsening used below in theorem 5.3 is applied only inside K_{H_α} , where α is an order unit, and it is never presented as a faithful image of Γ itself. The valuations discarded by that coarsening are exactly the ones recovered in section 5 by formal evaluation, not by a norm.

Tate’s original proof explicitly starts with a nontrivial real-valued absolute value and proves uniformization over any complete field with that absolute value [1]. Modern treatments include [2, 4]; the 2026 work of Eterović–Vermeulen concerns p -adic definability and Tate uniformization [6]. Higher-rank analytic geometry already exists, notably the Hahn analytification of Foster–Ranganathan [8]. Thus neither “higher-rank geometry” nor “non-Archimedean elliptic functions” is new here.

The potentially new package is more specific: an *if and only if* Hahn-support domain; a proof that this restricted domain nevertheless uniformizes every rational point; and exact compatibility with coarse reduction, scalar extension, and exponent-group arithmetic. No assertion of bibliographic uniqueness follows from the targeted search. Related formulations in formal, adic, or model-theoretic geometry may subsume parts of the result.

2 Hahn summation and the relevant order structure

2.1 Strong summability

A Hahn series is a formal expression

$$a = \sum_{\gamma \in \Gamma} a_\gamma t^\gamma, \quad a_\gamma \in \mathbb{C},$$

whose support $\text{supp}(a) = \{\gamma : a_\gamma \neq 0\}$ is well ordered in the increasing order. For $a \neq 0$, its valuation is the least support element. A family $(a_i)_{i \in I}$ is *strongly summable* when $\bigcup_i \text{supp}(a_i)$ is well ordered and each exponent occurs in the support of only finitely many a_i . Its sum is defined coefficientwise. Neither an order limit nor a limit in the full valuation topology is part of this definition.

Lemma 2.1 (Positive-support calculus). *Let $S \subseteq \Gamma_{>0}$ be well ordered. Its additive monoid S^* , including 0, is well ordered. Every element of Γ has only finitely many representations as a finite unordered sum of members of S , counting multiplicities. Consequently, for finitely many Hahn series $z_1, \dots, z_d$ of positive valuation, any formal power series $F \in \mathbb{C}[[Z_1, \dots, Z_d]]$ has a well-defined strong evaluation $F(z)$. Evaluation preserves sums, products, and composition when the inner formal series have zero constant coefficient.*

Proof. We recall the order-theoretic argument behind Neumann’s lemma [9]. The alphabet S , with its order, is well quasi-ordered. Higman’s lemma says that finite words over this alphabet are well quasi-ordered by order-preserving subsequence embedding with each letter allowed to increase [10]. If a word embeds into another, the sum of its letters does not exceed the sum of the other word. The inequality is strict if a letter increases or a positive letter is added. A descending sequence of sums would therefore contradict Higman’s lemma. An infinite collection of distinct nondecreasing words of the same sum would also contradict it: an embedding between two such words with equal sum forces the words to coincide. Sorting a word removes the finitely many permutations of each finite multiset. This proves both assertions.

The union of the finitely many positive supports $\text{supp}(z_i)$ is well ordered. Expanding all monomials in $F(z)$ produces exponents in S^* . The finiteness assertion bounds the word lengths and the decompositions at a fixed exponent, and the number of variable labels is finite. Hence every coefficient is a finite sum. The same argument for the jointly indexed families proves multiplication and the stated composition rule. This is the usual Hahn–Neumann calculus, not a new summation principle. $\square$

In particular, $(1 - z)^{-1} = \sum_{n \geq 0} z^n$ and $(1 + z)^r = \sum_{n \geq 0} \binom{r}{n} z^n$ for $r \in \mathbb{Q}$ and $v(z) > 0$. These identities hold even when $nv(z)$ is bounded above in Γ . A series with leading term ct^γ is inverted by factoring it as $ct^\gamma(1 + \varepsilon)$ and applying this rule to ε .

2.2 Normalization and two convex subgroups

Lemma 2.2 (Finite normalization). *For $\beta \in \Gamma$, the following are equivalent:*

- (i) $\beta \in H_\alpha$;
- (ii) *there is an integer m with $0 \leq \beta + m\alpha < \alpha$.*

Such an m is unique. If $\beta \notin H_\alpha$, then either $\beta > N\alpha$ for every $N \in \mathbb{N}$, or $\beta < -N\alpha$ for every $N \in \mathbb{N}$.

Proof. If $|\beta| \leq N\alpha$, the finite list of adjacent half-open intervals $[k\alpha, (k+1)\alpha)$ for sufficiently many integers k covers β . Their interiors and endpoint convention give existence and uniqueness. The converse is immediate. Convexity of H_α gives the last alternative, or it follows directly by separating the two signs. $\square$

There is another convex subgroup inside H_α :

$$H_\alpha^- = \{\gamma \in H_\alpha : n|\gamma| < \alpha \text{ for all } n \geq 1\}. \quad (13)$$

It consists of the values infinitesimal relative to α . The quotient H_α/H_α^- is nonzero and Archimedean. Indeed, for every positive $\gamma \notin H_\alpha^-$ some multiple of γ is at least α , and every element of H_α is bounded by a multiple of α . This proves the Archimedean comparison in the quotient. Hölder’s theorem embeds this quotient in $(\mathbb{R}, +)$; normalize the embedding so that α has image 1.

This construction is not new in the collection, and we do not present it as ours. By construction α is an *order unit* of H_α : its positive integer multiples are cofinal there. The companion report on entire Hahn functions states exactly the resulting passage as its `ent:lem:coarsening` [17]: for an order unit h of an exponent group, there is an order-preserving π to $\mathbb{R}$ with $\pi(h) = 1$ whose kernel is the convex subgroup of values infinitesimal relative to h , the formula $|a|_\pi = \exp(-\pi(v(a)))$ is a nontrivial real-valued non-Archimedean absolute value whose topology is the intrinsic valuation topology, and the Hahn field is complete for it. Applied with exponent group H_α and $h = \alpha$, that lemma is the statement we need, with kernel exactly (13). We cite it rather than claim it. One hypothesis must be carried across with it: that report’s standing assumption is that its exponent group is *divisible*, and *we do not assume Γ , or H_α , divisible anywhere in this paper* — section 10 depends on keeping nondivisible groups. Its proof of that lemma uses only the order-unit property

and no divisibility, so the statement does transfer; because the transfer is ours and not the cited report's, we retain the short direct argument in theorem 5.3 below, which uses no divisibility, rather than rest a step of the main theorem on a hypothesis we have dropped.

The two quotients serve different purposes. The quotient by H_α^- produces a rank-one valuation *inside* K_{H_α} . The quotient by H_α produces a coarser valuation on the *whole* field K_Γ , whose residue field is K_{H_α} . Interchanging them would destroy the proof.

3 The exact domain of the bilateral theta series

Theorem 3.1 (Theta domain). *The family in (6) is strongly summable exactly for $u \in U_q$. On that domain,*

$$\Theta_q(qu) = -u^{-1}\Theta_q(u). \quad (14)$$

Proof. Write $\beta = v(u)$. The leading exponent of its n th summand is

$$b_n = \frac{n(n-1)}{2}\alpha + n\beta. \quad (15)$$

Suppose first that $\beta \in H_\alpha$ and choose N with $|\beta| \leq N\alpha$. The differences $b_{n+1} - b_n = n\alpha + \beta$ show that the positive tail is eventually strictly increasing. Writing $n = -k$ gives the same property for the tail as $k \rightarrow +\infty$. Each tail eventually exceeds every prescribed element of H_α , by comparison with integer multiples of α . Together with the finitely many remaining indices, the set $B = \{b_n : n \in \mathbb{Z}\}$ is well ordered. Each value of b_n occurs at most twice: if $b_n = b_m$ and $n \neq m$, then $(n+m-1)\alpha + 2\beta = 0$.

Factor $q = c_q t^\alpha(1 + \varepsilon_q)$ and $u = c_u t^\beta(1 + \varepsilon_u)$ with positive-valuation errors. Let M be the monoid generated by their two supports. Every summand in (6) has support in $b_n + M$, by theorem 2.1, including negative powers of u . The sum of two well-ordered subsets of an ordered abelian group is well ordered, with finitely many pairs giving any fixed sum. Thus $B + M$ is well ordered. There are finitely many choices of n at a fixed leading value and finitely many decompositions in M . This proves strong summability of the original family.

If $\beta > N\alpha$ for all N , put $n = -k$. Then

$$b_{-(k+1)} - b_{-k} = (k+1)\alpha - \beta < 0.$$

These are actual nonzero leading coefficients, so their decreasing exponents already obstruct well-ordering of the union of supports. If $\beta < -N\alpha$ for all N , the sequence b_n decreases for $n \geq 0$. This proves necessity without assuming that possible cancellations between summands can repair an inadmissible family.

Finally, multiplication by q shifts the index in a strongly summable family. The coefficient equality $q^n(-1)^n q^{n(n-1)/2} = -(-1)^{n+1} q^{n(n+1)/2}$ proves (14). $\square$

Proposition 3.2 (Product and zeros). *For $0 \leq v(u) < \alpha$, the Jacobi product is a Hahn identity:*

$$\Theta_q(u) = (1-u) \prod_{n \geq 1} (1-q^n)(1-uq^n)(1-u^{-1}q^n). \quad (16)$$

The product means the strong sum over its finite subproducts after expansion. The zeros on U_q are exactly $q^{\mathbb{Z}}$, and all are simple in the local external-variable sense.

Proof. All of q , qu , and q/u have positive valuation. Their support monoid licenses the expanded product and its regrouping. The classical Jacobi triple-product identity, in this normalization, is an identity of the

corresponding formal Laurent families [14]; strong evaluation therefore yields (16). One can also compare coefficients of q after treating u as a Laurent indeterminate and then use the same support argument for specialization.

The product after the factor $1 - u$ is 1 plus a positive-valuation series, hence a unit. Therefore the only zero in the normalized annulus is $u = 1$. The functional equation and theorem 2.2 give all zeros. At $u = 1$ the derivative of the product is

$$\Theta'_q(1) = - \prod_{n \geq 1} (1 - q^n)^3 \neq 0.$$

Reindexing transports simplicity to every q^m . This is differentiation of a local power series in a variable; no derivation on the scalar field is being imposed. $\square$

4 Tate coordinates, support certificates, and precision

4.1 A form with positive tails

Put

$$f(z) = \frac{z}{(1-z)^2}, \quad g(z) = \frac{z^2}{(1-z)^3}, \quad h(z) = \frac{z}{(1-z)^3}.$$

The elementary identities $f(z^{-1}) = f(z)$ and $g(z^{-1}) = -h(z)$ give, after normalizing $0 \leq \beta = v(u) < \alpha$,

$$X_q(u) = f(u) + \sum_{n \geq 1} (f(q^n u) + f(q^n/u)) - 2s_1(q), \quad (17)$$

$$Y_q(u) = g(u) + \sum_{n \geq 1} (g(q^n u) - h(q^n/u)) + s_1(q). \quad (18)$$

The minus sign and the unsquared numerator in the last tail of (18) are important. In contrast, the numerator in the original bilateral formula (8) is squared.

At $v(u) = 0$ we do *not* expand the isolated expressions $f(u)$ and $g(u)$ geometrically in u . They are finite rational operations, defined unless $u = 1$. Their denominators may be arbitrarily small.

Theorem 4.1 (Coordinate domain and support certificate). *For $u \notin q^{\mathbb{Z}}$, the families in (7) and (8) are strongly summable exactly when $u \in U_q$. For $0 \leq v(u) < \alpha$ and $u \neq 1$, let*

$$S = \text{supp}(q) \cup \text{supp}(qu) \cup \text{supp}(q/u) \subseteq \Gamma_{>0}. \quad (19)$$

Then

$$\text{supp}(X_q(u) - f(u)), \text{supp}(Y_q(u) - g(u)) \subseteq S^* \setminus \{0\}. \quad (20)$$

The bilateral and positive-tail formulas agree.

Proof. The series $q^n u = q^{n-1}(qu)$ and $q^n/u = q^{n-1}(q/u)$ have positive valuation for $n \geq 1$. Expanding f, g, h in those arguments and expanding each term of s_1 gives exponents in $S^* \setminus \{0\}$. The full family, including n and every geometric-expansion index, is strongly summable: at a fixed exponent, the Neumann finiteness theorem bounds the total word length, hence also n and the geometric index. This proves (20) and justifies every regrouping in (17)–(18). Multiplication of u by a power of q merely reindexes the bilateral sums. Finite normalization therefore proves sufficiency on U_q .

For necessity, suppose first $\beta > N\alpha$ for every N . All $q^n u$ have positive valuation. The leading exponents in the X family are $\beta + n\alpha$ and in the Y family are $2\beta + 2n\alpha$. Both decrease strictly as $n \rightarrow -\infty$. If $\beta < -N\alpha$

for every N , all $q^n u$ have negative valuation. The rational identities above show that the leading exponents of both families are $-\beta - n\alpha$, with a nonzero sign in the Y case. These decrease as $n \rightarrow +\infty$. In either case the original families fail the well-ordering requirement. $\square$

Proposition 4.2 (Exact finite-tail bound). *For normalized u , let $X^{[N]}, Y^{[N]}$ retain the isolated rational terms and the terms $1 \leq n \leq N$ in the positive-tail formulas, and replace s_1 by $\sum_{n=1}^N nq^n/(1-q^n)$. Then*

$$v(X_q - X^{[N]}) \geq (N+1)\alpha - \beta, \quad v(Y_q - Y^{[N]}) \geq (N+1)\alpha - \beta. \quad (21)$$

Here $v(0) = +\infty$, so cancellations cause no exception.

Proof. Each omitted $f(q^n/u)$ or $h(q^n/u)$ has valuation $n\alpha - \beta$; the other omitted coordinate terms have at least that valuation. The omitted s_1 terms have valuation $n\alpha$. Strong summation and the valuation inequality give the bounds. $\square$

If a requested precision δ lies in H_α , a finite N certifies (21) beyond δ . If $\delta > N\alpha$ for all N , this certificate never reaches it. A valid exact Hahn sum has therefore not been converted into a convergence theorem at all scales.

4.2 The elliptic curve and its invariant differential

Proposition 4.3 (Discriminant and modular invariant). *The coefficients in (4) are Hahn evaluations of integer power series in q . The Weierstrass discriminant and j -invariant are*

$$\Delta(q) = q \prod_{n \geq 1} (1 - q^n)^{24}, \quad v(\Delta(q)) = \alpha, \quad (22)$$

$$j(q) = q^{-1} + 744 + 196884q + 21493760q^2 + \cdots, \quad v(j(q)) = -\alpha. \quad (23)$$

In particular E_q is a nonsingular elliptic curve over K_Γ .

Proof. The formulas for s_k are licensed by positive-support calculus. The integer divisibility of $(5n^3 + 7n^5)/12$ follows by checking its factors modulo 3 and 4, or from the standard Tate coefficient identity. With $a_1 = 1$, $a_2 = a_3 = 0$, the usual Weierstrass formulas give

$$b_2 = 1, \quad b_4 = 2a_4, \quad b_6 = 4a_6, \quad b_8 = a_6 - a_4^2, \quad c_4 = 1 - 48a_4.$$

The formal identities for Δ and c_4^3/Δ are classical [1, 2]. They are identities in integer formal power series (Laurent series for j), so positive-support evaluation proves them here. The product multiplying q is a unit of leading coefficient 1. Hence $\Delta \neq 0$, and the displayed valuations follow. $\square$

The coordinate functions satisfy the curve equation and the group law; a proof valid at every rank is given in section 7. The following local identity can already be checked term by term.

Proposition 4.4 (Differential and inversion identities). *On the local external-variable germs in U_q , away from poles,*

$$u \frac{dX_q}{du} = X_q + 2Y_q, \quad X_q(u^{-1}) = X_q(u), \quad Y_q(u^{-1}) = -X_q(u) - Y_q(u). \quad (24)$$

Consequently the invariant differential pulls back as

$$\Phi_q^* \left(\frac{dx}{2y + x} \right) = \frac{du}{u}. \quad (25)$$

The equation for differentials is interpreted in regular local charts also where the displayed affine denominator vanishes.

Proof. The rational identity $zf'(z) = f(z) + 2g(z)$ and the constants $-2s_1 + 2s_1 = 0$ prove the first formula. Differentiating the positive tails preserves their support estimates locally, so the interchange is licensed. The inversion formulas follow from $f(z^{-1}) = f(z)$ and $g(z^{-1}) = -f(z) - g(z)$, followed by reindexing. Equation (25) is the first identity rewritten as a differential. It extends in regular charts since an invariant differential has no pole on a smooth elliptic curve. $\square$

Nothing in this proposition says that a chosen scalar derivation $\partial : K \rightarrow K$ satisfies these formulas without the chain rule's q -dependence. In particular, it makes no assertion about the Berarducci–Mantova derivation of surreal scalars.

5 Coarse infinitesimals admit unrestricted formal evaluation

This section supplies the step that a bare appeal to “completeness of Hahn fields” would miss. We shall evaluate formal series whose coefficients have no uniform bound for the original valuation. A coarser valuation makes this possible.

5.1 The coarse residue field

Let $H \subseteq \Gamma$ be any convex subgroup. The quotient $\bar{\Gamma} = \Gamma/H$ is ordered. Define

$$w(a) = v(a) + H \quad (a \neq 0), \quad O_w = \{a : w(a) \geq 0\}, \quad \mathfrak{m}_w = \{a : w(a) > 0\}.$$

For $a \in O_w$, retaining exactly the terms with exponents in H gives a residue homomorphism

$$\text{res}_H : O_w \longrightarrow K_H, \quad \sum_{\gamma} a_{\gamma} t^{\gamma} \mapsto \sum_{\gamma \in H} a_{\gamma} t^{\gamma}. \quad (26)$$

Its kernel is $\mathfrak{m}_w$, and the inclusion $K_H \subseteq O_w$ is a coefficient-field section. No section of the ordered group quotient $\Gamma \rightarrow \Gamma/H$ is required.

To verify multiplicativity, observe that the support of an element of O_w contains only nonnegative quotient values. A product exponent can belong to H only if both summand exponents belong to H . Thus its H -part is exactly the product of the H -parts.

Lemma 5.1 (Unrestricted coarse formal evaluation). *Let $z_1, \dots, z_d \in \mathfrak{m}_w$. For every formal series*

$$F(Z) = \sum_{\nu \in \mathbb{N}^d} a_{\nu} Z^{\nu} \in K_H[[Z_1, \dots, Z_d]],$$

the family $(a_{\nu} z^{\nu})_{\nu}$ is strongly Hahn-summable in K_{Γ} . There is no boundedness assumption on $v(a_{\nu})$ in H . Evaluation is a K_H -algebra homomorphism. It commutes with formal composition when the inner series have zero constant coefficient.

Proof. Project the supports of the nonzero z_i into Γ/H and let S be their union. A monotone image of a well-ordered set is well ordered: the least preimage of any nonempty chosen subset maps to its least image. Since $w(z_i) > 0$, every element of S is strictly positive. Apply theorem 2.1 in the quotient group. The possible quotient values form the well-ordered monoid S^* . At a fixed quotient value $\bar{\gamma}$, there are only finitely

many decompositions into members of S , including word lengths. In particular, there are only finitely many multiindices ν that can contribute there.

For each of these finitely many choices, taking the indicated quotient fibers of the supports of the z_i and multiplying by a_ν involves ordinary finite Hahn products. These are well-defined and have well-ordered support within the fiber, with finite coefficient contributions. The finite union over the choices has the same properties. A subset of an ordered group with well-ordered quotient image and well-ordered support in every quotient fiber is well ordered: minimize first the quotient and then the fiber. Hence the union of all supports is well ordered in Γ , and each actual exponent receives finitely many contributions.

For addition and multiplication, use the same argument on the jointly indexed families. For composition, the zero-constant hypothesis ensures that at any bounded word length only finitely many formal substitutions can contribute. The quotient Neumann bound again supplies this word length bound. Thus the coefficientwise identities survive evaluation. $\square$

Example 5.2 (Why coefficient bounds are unnecessary). Let $\Gamma = \mathbb{Z} \oplus_{\text{lex}} \mathbb{Z}$, $H = \{0\} \oplus \mathbb{Z}$, and $z = t^{(1,0)}$. Then

$$\sum_{n \geq 0} t^{(0, -n^2)} z^n = \sum_{n \geq 0} t^{(n, -n^2)}$$

is strongly summable, although its coefficient valuations $(0, -n^2)$ are unbounded below in H . The first coordinate separates the degrees. This is precisely the type of separation used in the uniformization proof.

5.2 The rank-one field at the q -scale

Now set $H = H_\alpha$. Use the embedding $H/H_\alpha^- \hookrightarrow \mathbb{R}$ from section 2 to define a real-valued valuation v_1 on K_H , with $v_1(t^\alpha) = 1$.

Lemma 5.3 (Completeness of the residue workspace). *The field K_H , equipped with the absolute value $|a|_1 = e^{-v_1(a)}$, is complete. Every $q_0 \in K_H$ with $v(q_0) = \alpha$ has $|q_0|_1 < 1$.*

This is `ent:lem:coarsening` of [17], instantiated at the exponent group H_α with the order unit α , and is cited rather than claimed as new; see section 2. The proof below is recorded only because that report states its lemma under a standing divisibility hypothesis on the exponent group which this paper does not make, and because the package should read standalone. It is a second proof of a statement already in the collection, not an additional result.

Proof. Let (a_n) be a Cauchy sequence. Choose a subsequence (a_{n_k}) whose successive differences $d_k = a_{n_{k+1}} - a_{n_k}$ satisfy $v_1(d_k) > k$. Then $v(d_k) > k\alpha$. The sequence $k\alpha$ is cofinal in H : for $\gamma \in H$, choose an integer bound for $|\gamma|$ and then a strictly larger integer.

It follows that (d_k) is strongly summable in K_H . Indeed, below any fixed exponent only finitely many of its supports can occur; any hypothetical descending sequence of support exponents would therefore lie in a finite union of well-ordered supports. The strong sum $a_{n_1} + \sum_k d_k$ is the v_1 -limit of the subsequence. The original Cauchy sequence has the same limit. The assertion about q_0 is $v_1(q_0) = 1$. $\square$

The coefficient field of this rank-one valuation need not be $\mathbb{C}$; it can itself retain infinitesimal scales. Tate's theorem does not require it to be $\mathbb{C}$ or require the value group to be discrete. Its complete-field version is exactly the classical input needed here.

6 Formal germs and the passage from rank one to arbitrary rank

We record carefully how classical analytic identities can be used without illegally specializing a non-summable family.

6.1 The classical input, stated with its hypotheses

Theorem 6.1 (Tate; external classical input). *Let F be complete for a nontrivial real-valued non-Archimedean absolute value, and let $q_0 \in F$ satisfy $0 < |q_0| < 1$. The rational bilateral series (7)–(8), interpreted by their convergent tails, define an analytic group homomorphism*

$$\phi_{q_0} : F^\times \longrightarrow E_{q_0}(F)$$

which is onto and has kernel $q_0^{\mathbb{Z}}$. It extends analytically through those parameters as a map to the projective curve. The same formulas are locally analytic in the parameter q_0 as well as in u .

The surjective group statement is Tate’s Theorem 1 [1]. Local analyticity in both variables follows directly from the rational tails: on sufficiently small neighborhoods of any non-pole pair, the finitely many relevant denominators remain nonzero and the remaining terms tend uniformly to zero geometrically. At a pole one uses the projective parameter below. This last observation requires no new uniformization theorem.

The proof of theorem 1.2 uses theorem 6.1 as an input, not as a theorem claimed to have been independently reproved here. All additional all-rank summability and lifting steps are supplied.

6.2 The formal neighborhood of the identity

Lemma 6.2 (A universal formal parameter). *With an external variable h , the formal expression*

$$B(q, h) = -\frac{X_q(1+h)}{Y_q(1+h)} \tag{27}$$

after cancellation of its poles belongs to $h\mathbb{C}[[q, h]]$ and has linear coefficient 1 in h :

$$B(q, h) = h + h^2 C(q, h), \quad C \in \mathbb{C}[[q, h]]. \tag{28}$$

It has a unique inverse under composition in the second variable, $A(q, z) = z + z^2 D(q, z) \in z\mathbb{C}[[q, z]]$. For any $v(q) > 0$ and $v(h) > 0$, both series evaluate strongly, and

$$v(B(q, h)) = v(h), \quad \text{lc}(B(q, h)) = \text{lc}(h). \tag{29}$$

The germ $h \mapsto \Phi_q(1+h)$ identifies the multiplicative formal group with the formal group of E_q at O .

Proof. Expand the positive tails with $u = 1 + h$, treating q, h as formal variables. They belong to $q\mathbb{C}[[q, h]]$. Hence

$$\begin{aligned} h^2 X_q(1+h) &= 1 + h + h^2 R_X(q, h), \\ h^3 Y_q(1+h) &= -(1+h)^2 + h^3 R_Y(q, h), \end{aligned}$$

where $R_X, R_Y \in q\mathbb{C}[[q, h]]$. The denominator obtained after cancelling poles has constant coefficient 1. This proves (28), including the exact coefficient 1 for every q . Formal coefficient recursion gives the inverse A , since this linear coefficient is a unit.

Positive-support calculus evaluates B , A , and their composition identities at the specified arguments. The remainder $h^2C(q, h)$ has valuation at least $2v(h) > v(h)$, which proves (29).

The formal-group identity is a universal identity of formal series in q, h_1, h_2 . For ordinary complex q and sufficiently small h_1, h_2 it follows from classical multiplicative uniformization. Uniqueness of Taylor coefficients gives the formal identity; alternatively it is the standard formal Tate-curve construction [1, 5]. Strong evaluation of the jointly indexed positive-support family proves it in K_Γ . $\square$

Here is a direct explanation of why this formal germ parametrizes all points in a coarse identity fiber, rather than just some of them. Use projective coordinates $[X : Y : Z]$ near $O = [0 : 1 : 0]$ and set

$$z = -X/Y = -x/y, \quad r = -Z/Y = -1/y.$$

The curve equation becomes

$$r(1 - z) = z^3 + a_4 z r^2 + a_6 r^3. \quad (30)$$

It has a unique solution $r \in z^3\mathbb{C}[[q, z]]$ by coefficient recursion. For $z \in \mathfrak{m}_w$, positive-support evaluation gives such a solution. Uniqueness also holds among actual $r \in \mathfrak{m}_w$: subtracting two solutions factors their difference times

$$1 - z - a_4 z(r_1 + r_2) - a_6(r_1^2 + r_1 r_2 + r_2^2),$$

which is a w -unit because its residue is 1. Thus two points reducing to O and having the same z coincide. Evaluation of the inverse $A(q, z)$ supplies the unique corresponding $h \in \mathfrak{m}_w$.

6.3 Taylor prolongation through coarse infinitesimals

Let $w = v \bmod H_\alpha$, set $F = K_{H_\alpha}$, and put $q_0 = \text{res}_{H_\alpha}(q)$. Then

$$q = q_0(1 + \rho), \quad \rho \in \mathfrak{m}_w, \quad u = u_0(1 + \sigma), \quad \sigma \in \mathfrak{m}_w, \quad u_0 = \text{res}_{H_\alpha}(u) \in F^\times$$

for every $u \in U_q$.

Lemma 6.3 (Agreement of Taylor and Hahn evaluation). *The classical local Taylor germ of the projective map $(Q, U) \mapsto \phi_Q(U)$ at (q_0, u_0) , evaluated at (ρ, σ) by theorem 5.1, is well defined and equals $\Phi_q(u)$ as defined by the literal Hahn formulas. This remains true when $u_0 \in q_0^{\mathbb{Z}}$, using a projective local chart.*

Proof. First suppose $u_0 \notin q_0^{\mathbb{Z}}$. Replace (u, u_0) by $(uq^m, u_0q_0^m)$ so that $0 \leq v(u_0) < \alpha$. This is a finite reindexing of the bilateral formulas, and its relative error is still in $\mathfrak{m}_w$.

The isolated rational terms have nonzero constant denominators in F . Their Taylor series in ρ, σ evaluate to the actual rational terms by theorem 5.1. For the n th positive rational tail, expand first in the relative variables. At every fixed Taylor multiindex, the coefficient lies in F and its fine valuation tends to $+\infty$ cofinally in H_α as $n \rightarrow \infty$. More explicitly, after discarding finitely many n , the denominators $1 - q_0^n u_0$ and $1 - q_0^n / u_0$ are fine units, and a coefficient of any fixed relative Taylor degree has valuation at least $n\alpha - v(u_0)$, or a larger positive multiple for the g -tail. Differentiating relative powers only contributes ordinary integer coefficients and additional factors of these small rational arguments. The finitely many discarded denominators affect no tail conclusion.

It follows that the sum over n defining every Taylor coefficient is a strong sum in F and agrees with its rank-one analytic sum, by theorem 5.3. Next fix a quotient exponent for w . The quotient Neumann bound in theorem 5.1 permits only finitely many Taylor multiindices and quotient-support decompositions at this value. For each such choice, the remaining sum over n is the cofinally bounded sum just described. Thus the

family jointly indexed by Taylor degree, tail index, and support decompositions is strongly summable. Fubini regrouping for this joint family proves that Taylor evaluation and the original Hahn-tail evaluation coincide. The s_1 family has the same, simpler estimates.

If $u_0 = q_0^m$, divide u by q^m . The result is $1 + h$ with $h \in \mathfrak{m}_w$. Apply theorem 6.2 and the regular projective chart (30). The poles of X, Y disappear in this chart, and the same formal expressions are the Taylor germs of the classical projective map. This proves the claim, including $h = 0$. $\square$

Remark 6.4 (The interchange that must not be omitted). Formal coefficients in F can have arbitrarily negative fine valuations, as theorem 5.2 shows. Ordinary positive-valuation substitution over $\mathbb{C}$ alone does not justify their evaluation. Also, a sequence cofinal in H_α need not be cofinal in Γ . The proof above uses two separate finiteness statements: quotient Neumann finiteness controls the coarse Taylor degrees, while cofinality inside H_α controls the rational tail index. Neither statement substitutes for the other.

7 Proof of arbitrary-rank uniformization

7.1 The curve equation and the group law

Take $F = K_{H_\alpha}$, with its complete rank-one coarsening, as in theorem 5.3. Since $v(q_0) = \alpha$, the classical theorem applies to q_0 . The projective Taylor germs in theorem 6.3 satisfy the curve equation: it is an identity of classical analytic germs, hence an identity of their Taylor series. Evaluation preserves these polynomial identities. Thus Φ_q maps into $E_q(K)$.

The same argument proves

$$\Phi_q(uv) = \Phi_q(u) + \Phi_q(v) \quad (u, v \in U_q). \quad (31)$$

Here one must use regular local charts on the elliptic curve, not the chord-slope formula at a vanishing denominator. The group operation on a smooth elliptic curve is a regular morphism. At the residue points $\phi_{q_0}(u_0), \phi_{q_0}(v_0)$ and their sum, choose affine neighborhoods for which it is given by rational expressions with nonzero residue denominators. Its local Taylor identity with $\phi_Q(UV)$ follows from theorem 6.1. All denominators remain w -units after coarse evaluation, and theorem 5.1 preserves the identity. At the identity, use (30). This includes $u_0v_0 \in q_0^{\mathbb{Z}}$, opposite points, equal points, and every other case where a particular affine addition formula would be undefined. By theorem 6.3, these evaluated germs are the Hahn-defined map, proving (31) everywhere on U_q .

For clarity, this is not an appeal to a new “identity principle on all surcomplex numbers.” It is an ordinary identity of rank-one analytic germs, followed by an explicitly proved algebra homomorphism on their formal Taylor rings.

7.2 Good reduction for the coarse valuation

By (22), $v(\Delta(q)) = \alpha \in H_\alpha$, so $w(\Delta(q)) = 0$. All coefficients of the projective Weierstrass equation belong to $\mathcal{O}_w$, and its discriminant is a unit there. It is therefore a smooth proper elliptic curve over $\mathcal{O}_w$, with residue curve E_{q_0} over F . The identification of the residue curve is the statement $\text{res}_{H_\alpha}(a_k(q)) = a_k(q_0)$ for $k = 4, 6$. It holds because res_{H_α} is a ring homomorphism on $\mathcal{O}_w$ that is computed coefficientwise, hence commutes with the strong sums (3): each term $n^k q^n / (1 - q^n)$ lies in $\mathcal{O}_w$, since $v(q^n) = n\alpha > 0$ forces $v(1 - q^n) = 0$, and $\text{res}_{H_\alpha}(q) = q_0$.

Every K -point has a well-defined reduction to that curve. In the present projective model, one can see existence directly: scale its three homogeneous coordinates so that their minimum w -value is zero, and reduce

the resulting integral coordinates. The equation survives reduction, and at least one reduced coordinate is nonzero. Smooth properness also implies that reduction respects the group law. We write

$$\text{red} : E_q(K) \longrightarrow E_{q_0}(F), \quad E_1(K) = \ker(\text{red}).$$

The reduction compatibility supplied by theorem 6.3 is

$$\text{red}(\Phi_q(u)) = \phi_{q_0}(u_0). \quad (32)$$

Proposition 7.1 (The whole coarse identity fiber). *The homomorphism $h \mapsto \Phi_q(1+h)$ induces an isomorphism*

$$1 + \mathfrak{m}_w \simeq E_1(K). \quad (33)$$

For the local coordinate $z = -x/y$, it has the explicit inverse $h = A(q, z)$ and satisfies $v(z) = v(h)$ for $h \neq 0$.

Proof. A point reducing to $O = [0 : 1 : 0]$ lies in the chart of (30), with $z, r \in \mathfrak{m}_w$. The discussion following theorem 6.2 proves that z uniquely determines the point and that the inverse formal parameter supplies a unique $h \in \mathfrak{m}_w$. Formal composition identities show that these maps are mutual inverses. The formal-group identity gives the group law; the valuation assertion is (29). $\square$

7.3 Surjectivity

Let $P \in E_q(K)$ be arbitrary, and reduce it to $P_0 \in E_{q_0}(F)$. By the classical Tate theorem there is $u_0 \in F^\times$ with $\phi_{q_0}(u_0) = P_0$. Regard u_0 as an element of K . Its valuation belongs to H_α , so $\Phi_q(u_0)$ is defined. By (32) the point

$$P - \Phi_q(u_0)$$

belongs to $E_1(K)$. By theorem 7.1 it equals $\Phi_q(1+h)$ for a unique $h \in \mathfrak{m}_w$. The group law now gives

$$P = \Phi_q(u_0(1+h)).$$

The product parameter is in U_q . This proves surjectivity, including points with coordinates outside every scale bounded by powers of q .

When $H_\alpha = \Gamma$, the valuation w is trivial and $\mathfrak{m}_w = 0$. The proof reduces to the classical theorem on $K_H = K$ with its rank-one coarsening. No separate nontrivial-coarse-valuation hypothesis was smuggled into the argument.

7.4 The kernel

Suppose $\Phi_q(u) = O$. Reducing and using the classical kernel shows $u_0 = q_0^m$ for an integer m . Therefore $u/q^m \in 1 + \mathfrak{m}_w$. Reindexing already gives $\Phi_q(q^m) = O$; the group law and theorem 7.1 imply $u/q^m = 1$. Conversely every q^m is mapped to O by definition. Thus $\ker \Phi_q = q^\mathbb{Z}$. Together with the support theorems and surjectivity, this completes the proof of theorem 1.2.

7.5 The exact reduction diagram

The proof gives more than an abstract isomorphism.

Corollary 7.2 (Reduction and infinitesimal lifting). *There is a commutative diagram of exact sequences*

$$\begin{array}{ccccccc}
 1 & \longrightarrow & 1 + \mathfrak{m}_w & \longrightarrow & U_q/q^{\mathbb{Z}} & \xrightarrow{\text{res}_H} & F^\times/q_0^{\mathbb{Z}} \longrightarrow 1 \\
 & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\
 0 & \longrightarrow & E_1(K) & \longrightarrow & E_q(K) & \xrightarrow{\text{red}} & E_{q_0}(F) \longrightarrow 0.
 \end{array}$$

The left-hand isomorphism is given by a support-controlled universal formal series and preserves the original valuation of the local identity parameter.

Proof. Surjectivity of the top residue map follows by taking representatives in F . Its kernel consists of those classes whose residue is a power of q_0 ; division by the corresponding power of q gives a unique representative in $1 + \mathfrak{m}_w$. Uniqueness uses $v(q_0) > 0$. The lower row and all vertical maps were established above. $\square$

8 Scalar extension and the valuation-circle quotient

8.1 Strong functoriality

Theorem 8.1 (Exponent embeddings preserve uniformization). *Let $\iota : \Gamma \hookrightarrow \Gamma'$ be an ordered group embedding. The coefficient-preserving map*

$$\iota_* : K_\Gamma \hookrightarrow K_{\Gamma'}, \quad \sum a_\gamma t^\gamma \mapsto \sum a_\gamma t^{\iota(\gamma)}$$

is a field embedding preserving strong sums. If $q' = \iota_(q)$, then*

$$\iota_*(\Phi_q(u)) = \Phi_{q'}(\iota_*(u)) \quad (u \in U_q), \quad (34)$$

with the corresponding identities for Θ, X, Y where defined. Moreover

$$\iota^{-1}(H_{\iota(\alpha)}) = H_\alpha. \quad (35)$$

In the larger field the full admissible domain is $v^{-1}(H_{\iota(\alpha)})$, not in general all of $K_{\Gamma'}^\times$.

Proof. An order embedding preserves and reflects well-ordering and preserves finite coefficient contributions. It therefore preserves the defining Hahn operations and every strongly summable family. Apply this to the bilateral formulas, or to their positive-tail forms. The point at infinity and the powers of q are also preserved. Finally, inequalities $|\iota(\gamma)| \leq N\iota(\alpha)$ are equivalent to $|\gamma| \leq N\alpha$, proving (35). The exact-domain theorems over Γ' give the last assertion. $\square$

This theorem has a useful asymmetric meaning. An old admissible parameter never becomes inadmissible under an ordered exponent embedding. But the assertion “every nonzero parameter is admissible” may fail after enlargement: new valuations can lie outside the old q -scale. The curve acquires new points, yet its correct, enlarged admissible subgroup still reaches all of them.

Example 8.2 (A rank-one base and a rank-two extension). Embed $\mathbb{R}$ as the second summand in $\Gamma' = \mathbb{R} \oplus_{\text{lex}} \mathbb{R}$, and take $q = t^{(0,1)}$. Over $\mathbb{C}((t^{\mathbb{R}}))$, every nonzero parameter is admitted. Over $K_{\Gamma'}$, a parameter of valuation $(1,0)$ is not admitted. A parameter $1 + t^{(1,0)}$ is admitted and yields a new point reducing to O for the coarse first-coordinate valuation. Thus excluding a new multiplicative scale does not exclude its infinitesimal displacement from a nonzero old parameter.

8.2 A canonical circularly ordered value quotient

Let $O_v^\times = \{u \in K^\times : v(u) = 0\}$ be the units of the original Hahn valuation ring.

Theorem 8.3 (The valuation quotient). *There is a canonical surjective group homomorphism*

$$\text{trop}_q : E_q(K) \longrightarrow H_\alpha/\mathbb{Z}\alpha, \quad \text{trop}_q(\Phi_q(u)) = v(u) + \mathbb{Z}\alpha, \quad (36)$$

and an exact sequence

$$1 \longrightarrow O_v^\times \xrightarrow{\Phi_q} E_q(K) \xrightarrow{\text{trop}_q} H_\alpha/\mathbb{Z}\alpha \longrightarrow 0. \quad (37)$$

Every value class has a unique representative in $[0, \alpha)$. The ordering of that interval defines a translation-invariant circular order on the quotient, with addition followed by reduction modulo α .

Proof. Two parameters represent the same point exactly when their quotient is a power of q , so their valuations differ by an integer multiple of α . This proves well-definedness. Every $\gamma \in H_\alpha$ is the valuation of t^γ , proving surjectivity. If $v(u) = m\alpha$, the same point is represented by u/q^m , which is a valuation-zero unit. Conversely units map to zero in the value quotient. Their map into $E_q(K)$ is injective, since a power of q has valuation zero only when it is 1. Finite normalization gives the unique representative. The usual cyclic order of three representatives in a fundamental half-open interval is independent of simultaneous translation, by cutting the interval at the translating point. $\square$

There is a second exact sequence of additive groups

$$0 \longrightarrow H_\alpha^- \longrightarrow H_\alpha/\mathbb{Z}\alpha \longrightarrow (H_\alpha/H_\alpha^-)/\mathbb{Z}\tilde{\alpha} \longrightarrow 0. \quad (38)$$

Indeed $H_\alpha^- \cap \mathbb{Z}\alpha = 0$, and the kernel computation is immediate. This separates infinitesimal value directions from the rank-one circular quotient.

When the relevant rank-one value group is all of $\mathbb{R}$, this value quotient is the familiar real circle of length $v_1(q)$. Over a complete algebraically closed rank-one field it agrees with the value-circle description of the Tate curve skeleton [7]. At arbitrary rank, (37) is an *algebraic and circular-order* assertion. It does not claim the existence of an ordinary real-metric Berkovich skeleton with all of H_α as its distance set. Higher-rank analytifications require their own definitions [8].

9 Recovering the Tate parameter without adding scales

9.1 The inverse modular series

Write $J = 1/j$. The formal series

$$J(q) = \frac{\Delta(q)}{c_4(q)^3} \in q\mathbb{Z}[[q]]$$

has linear coefficient 1, hence a unique compositional inverse $Q(J) \in J\mathbb{Z}[[J]]$. Its first terms are

$$Q(J) = J + 744J^2 + 750420J^3 + 872769632J^4 + 1102652742882J^5 + \cdots. \quad (39)$$

The notation Q here denotes a formal series, not the rational field.

Theorem 9.1 (Unique, support-preserving recovery). *Given $j \in K_\Gamma$ with $v(j) < 0$, there is a unique $q \in K_\Gamma$ with $v(q) > 0$ and $j(q) = j$. It is $q = Q(j^{-1})$ and satisfies $v(q) = -v(j)$. If $G(a)$ denotes the additive subgroup of Γ generated by $\text{supp}(a)$, then*

$$G(q) = G(j). \quad (40)$$

More precisely,

$$\text{supp}(q) \subseteq (\text{supp}(j^{-1}))^* \setminus \{0\}. \quad (41)$$

Proof. Since $v(j^{-1}) > 0$, positive-support calculus evaluates Q and both formal inverse identities. Thus $J(q) = j^{-1}$ and $j(q) = j$. The leading coefficient of Q is 1, so $v(q) = v(j^{-1})$. If another positive-valuation q' gives $j(q') = j$, evaluating the identity $Q(J(q')) = q'$ proves equality. This argument does not choose a branch of a transcendental function.

Formula (41) follows directly from the positive powers of j^{-1} . Inversion of a Hahn series uses only the additive subgroup of its support, so $G(j^{-1}) \subseteq G(j)$. Therefore $G(q) \subseteq G(j)$. Conversely $j(q) = q^{-1}$ times a power series in q with nonzero constant coefficient. Inversion and strong positive evaluation use only $G(q)$, giving $G(j) \subseteq G(q)$. $\square$

The inverse modular expansion is classical. The support equality is the useful extra statement in a symbolic or nondivisible Hahn workspace: one need not pass to the divisible hull merely to recover the elliptic period parameter.

9.2 Which elliptic curves this covers

Corollary 9.2 (Negative- j uniformization). *Assume Γ is divisible. Every elliptic curve E/K_Γ with $v(j(E)) < 0$ is K_Γ -isomorphic to E_q , for the unique parameter of theorem 9.1. After any such choice of isomorphism, $E(K_\Gamma)$ is parametrized by $U_q/q^\mathbb{Z}$.*

Proof. A Hahn field with algebraically closed coefficient field of characteristic zero and divisible exponent group is algebraically closed [11]. Elliptic curves over an algebraically closed field of characteristic zero with the same j -invariant are isomorphic [3]. Apply this to E and E_q . $\square$

The parameter q is canonical, but an isomorphism $E \simeq E_q$ need not be canonical. The hypothesis on j excludes $j = 0$ and 1728, but does not remove the usual sign ambiguity of an elliptic isomorphism. For nondivisible Γ , theorem 9.1 still holds, while a curve with the same j can be a nontrivial twist; we do not silently identify all such twists with the split Tate curve over the original field. Curves with $v(j) \geq 0$ are outside this positive- q theorem.

10 Torsion fields and exact exponent ramification

This section retains nondivisible exponent groups deliberately. Replacing Γ by its divisible hull at the start would make every degree computed here equal to one after base extension.

10.1 Roots of a Hahn parameter

Lemma 10.1 (A single radical extension). *Let $n \geq 2$, $q \in K_\Gamma$ with $v(q) = \alpha > 0$, and $\Lambda = \Gamma + \mathbb{Z}\alpha/n$. Then for any choice of $q^{1/n}$ in $K_{\Gamma \otimes \mathbb{Q}}$,*

$$K_\Gamma(q^{1/n}) = K_\Lambda, \quad [K_\Lambda : K_\Gamma] = [\Lambda : \Gamma] = \text{ord}(\alpha + n\Gamma). \quad (42)$$

The extension is cyclic, with Galois group $\text{Hom}(\Lambda/\Gamma, \mathbb{C}^\times)$; it has unchanged residue field $\mathbb{C}$ and value group Λ .

Proof. Write $q = ct^\alpha(1 + \varepsilon)$, $c \in \mathbb{C}^\times$, $v(\varepsilon) > 0$. Choose $c^{1/n} \in \mathbb{C}$. The binomial series gives $b = (1 + \varepsilon)^{1/n} \in K_\Gamma$, with $b^n = 1 + \varepsilon$. Consequently $q^{1/n} = c^{1/n}t^{\alpha/n}b$ up to a root of unity already in $\mathbb{C}$, so adjoining $q^{1/n}$ is the same as adjoining $t^{\alpha/n}$.

The group Λ/Γ is finite cyclic. If its order is d , partition any Hahn support in Λ among its d cosets. Factoring a fixed monomial from each part gives a finite sum of d monomials with coefficients in K_Γ . These monomials are linearly independent over K_Γ , since their supports lie in disjoint cosets. Thus the entire Hahn field K_Λ , not just a smaller subfield with finite supports, is a degree- d extension generated by $t^{\alpha/n}$.

Multiplying the monomial of exponent λ by a character of $\lambda + \Gamma$ defines a field automorphism, preserving strong sums. There are d such characters because $\mathbb{C}$ contains all roots of unity, and they give all automorphisms. The coefficient residue field remains $\mathbb{C}$ and the ramification index is d , equal to the field degree. Finally d is the least positive integer with $d\alpha \in n\Gamma$, which is the displayed order in $\Gamma/n\Gamma$. $\square$

10.2 A primitive torsion point

Theorem 10.2 (The complete torsion field). *Under the hypotheses of theorem 10.1, the n^2 points*

$$\Phi_q(\zeta_n^a q^{b/n}), \quad 0 \leq a, b < n, \quad (43)$$

are precisely $E_q[n]$. Their field of definition is K_Λ , and already

$$K_\Gamma(\Phi_q(q^{1/n})) = K_\Lambda. \quad (44)$$

Here adjoining a point means adjoining its affine coordinates.

Proof. Every parameter in (43) lies in the admissible group of K_Λ , since its valuation is $b\alpha/n$. Its n th power is q^b , so the group law makes its image n -torsion. If two of the points coincide, injectivity modulo $q^{\mathbb{Z}}$ gives

$$\zeta_n^{a-a'} q^{(b-b')/n} = q^m$$

for some integer m . Comparing valuations gives $b - b' = nm$; the chosen ranges force $m = 0$ and $b = b'$, and then $a = a'$. Thus there are n^2 distinct points. In characteristic zero an elliptic curve has exactly n^2 geometric n -torsion points, proving completeness of the list. They all have coordinates in K_Λ by strong evaluation.

To prove the stronger primitive assertion, let $L = K_\Gamma(q^{1/n}) = K_\Lambda$ and let $\sigma \in \text{Gal}(L/K_\Gamma)$ fix the point $P = \Phi_q(q^{1/n})$. The character description in theorem 10.1 implies that σ preserves every strong sum, so it commutes with the coordinate formulas. Therefore

$$\Phi_q(\sigma(q^{1/n})) = \Phi_q(q^{1/n}).$$

The quotient $\sigma(q^{1/n})/q^{1/n}$ is a root of unity. By the uniformization kernel it must also be a power of q . A root of unity has valuation zero, whereas $v(q^m) = m\alpha$; hence that power is 1. Thus σ fixes $q^{1/n}$ and is the identity. The stabilizer of P is trivial, so Galois theory gives (44). The full torsion field contains this primitive point field and is contained in L , proving equality. $\square$

Corollary 10.3 (A finite-lattice degree formula). *If $\Gamma \cong \mathbb{Z}^r$ as an abelian group, with any compatible total order, and α has coordinates $(a_1, \dots, a_r)$, then*

$$[K_\Gamma(E_q[n]) : K_\Gamma] = \frac{n}{\gcd(n, a_1, \dots, a_r)}. \quad (45)$$

Proof. The order of $(a_1, \dots, a_r)$ in $(\mathbb{Z}/n\mathbb{Z})^r$ is the least $d > 0$ for which n divides every da_i . This is the displayed integer. $\square$

For example, $n = 12$ and $\alpha = (6, 4)$ give degree 6; $n = 4$ and $\alpha = (0, 6)$ give degree 2 in the lexicographic group $\mathbb{Z}^2$. The entire unit part of q can have infinite support and does not alter these degrees. It is removed by a binomial root already in the base field. Over $K_{\mathbb{Q}}$ every n th root scale is already available, so the corresponding degrees are one. This explains why scalar extension can erase arithmetic information while preserving all uniformization identities.

11 Transport to surreal and surcomplex numbers

11.1 Set-sized normal-form workspaces

The Conway normal form writes a surreal as a reverse-well-ordered sum of real multiples of monomials ω^γ . With the convention $t^\gamma = \omega^{-\gamma}$, increasing well-ordered Hahn supports correspond to these normal forms. Combining the real and imaginary normal forms produces complex coefficients and gives the surcomplex version [12, 13].

If a set of surcomplex parameters and point coordinates is specified, the union of all of its normal-form exponent supports is a set. Its additive subgroup $\Gamma \subseteq (\mathbf{No}, +)$ is a set as well. The associated normal-form map embeds K_Γ into $\mathbf{No}[i]$ and contains all the specified elements. Enlarge to the divisible hull only when algebraic closedness is actually needed.

The proof of theorem 1.2 works in this set-sized workspace. If a larger workspace is chosen, theorem 8.1 shows that the formulas and their values agree. There is therefore no choice-dependent global summation rule and no need to regard a proper class as a set.

Theorem 11.1 (Surcomplex uniformization). *For every $q \in \mathbf{No}[i]^\times$ with $\alpha = v(q) > 0$, the literal Hahn Tate map has exactly the admissible domain in (12) and induces the group isomorphism stated there. This domain is always a proper subclass of $\mathbf{No}[i]^\times$. For every elliptic curve $E/\mathbf{No}[i]$ with $v(j(E)) < 0$, the unique parameter $q = Q(j(E)^{-1})$ yields an isomorphic Tate curve.*

Proof. The domain test for one parameter involves only α , its valuation, and ordinary integer inequalities. Its truth is unchanged in any workspace containing those data. The same holds for all the strong families in the formulas. For surjectivity, put q and a given point's coordinates in one workspace and apply theorem 1.2 there. For the kernel and the group law, place the finitely many relevant parameters in a common workspace. These are instances of theorem 8.1.

To see properness, take a positive surreal value $\gamma = \omega\alpha$. It exceeds $N\alpha$ for every ordinary N . The monomial $u = \omega^{-\gamma}$ lies outside the domain, by the exact support theorem.

For the final assertion put the Weierstrass coefficients of E in a workspace and take its divisible hull. The algebraically closed Hahn field there contains an isomorphism supplied by theorem 9.2. Its image in $\mathbf{No}[i]$ proves the assertion. The parameter obtained by the inverse series is independent of the workspace, again by strong functoriality. $\square$

11.2 A forbidden monomial and an admitted fine displacement

Example 11.2 (Two ways to use $\omega^{-\omega}$). Let $q = \omega^{-1}$, so $\alpha = 1$, and put $\varepsilon = \omega^{-\omega}$. The parameter $u = \varepsilon$ is inadmissible: $v(u) = \omega > N$ for all ordinary N . For the theta family the exponents along $n = -k$ are

$$\frac{k(k+1)}{2} - k\omega,$$

a strictly decreasing sequence. For the X family they include $\omega - k$, again strictly decreasing. No cancellation argument can make either original family strongly summable.

In contrast $u = 1 + \varepsilon$ has valuation zero and is admissible. The point has

$$\begin{aligned} X_q(1 + \varepsilon) &= \varepsilon^{-2} + \varepsilon^{-1} + \text{a positive-valuation tail}, \\ Y_q(1 + \varepsilon) &= -\varepsilon^{-3} - 2\varepsilon^{-2} - \varepsilon^{-1} + \text{a positive-valuation tail}, \\ -X_q(1 + \varepsilon)/Y_q(1 + \varepsilon) &= \varepsilon + O(\varepsilon^2). \end{aligned}$$

The last O means valuation at least 2ω , not an ordinary real asymptotic estimate. Thus an infinitesimal displacement far finer than the q -scale is retained exactly, and its image has correspondingly large elliptic coordinates.

11.3 Why fine-topological convergence is not being used

For $q = \omega^{-1}$, the powers q^n tend to zero in a rank-one workspace such as $\mathbb{C}((t^{\mathbb{R}}))$. They do not tend to zero when neighborhoods can test against every surreal positive scale: none of the ordinary q^n is smaller than $\omega^{-\omega}$. The repository already emphasizes this distinction between its fixed rank-one analysis and full fine topology [16].

The present proof uses ordinary convergence only in the named complete rank-one coarsening of K_{H_α} . Everywhere else it uses strong Hahn summation or finite algebra. Its finite-tail bound is a statement in the ordered value group; it is not relabeled as an all-scale limit. Neither a global surcomplex exponential nor a value of $\sin \omega$ is required. Multiplicative periods $q^{\mathbb{Z}}$ replace those additive analytic constructions.

12 Exact computation, proof dependencies, and implementation

12.1 Finite checks included with this manuscript

The accompanying program code/`check_identities.py` uses only exact rational symbolic arithmetic. Its recorded output is in `data/verification.json` and `data/verification.log`. The source package recorded a run on Python 3.13.5 and SymPy 1.14.0; the run recorded in the included data files was made when this report was placed in the collection, on Python 3.14.4 and SymPy 1.14.0, and it reproduced every recorded value except the Python version string itself. The JSON records the actual runtime version and is authoritative if the script is rerun elsewhere. The checks do not require a numerical value of q or a realization of an arbitrary ordered group.

For clarity, the coordinate coefficients used by the script are

$$[q^d]X_q(u) = \sum_{m|d} m(u^m + u^{-m} - 2), \quad d \geq 1, \tag{46}$$

$$[q^d]Y_q(u) = \sum_{m|d} \left(\frac{m(m-1)}{2} u^m - \frac{m(m+1)}{2} u^{-m} + m \right), \quad d \geq 1. \tag{47}$$

The degree-zero coefficients are $f(u)$ and $g(u)$. These identities come directly from the positive rational tails; they are also recorded in the contemporary Tate formulas of [6]. The script checks the curve equation, (24), and the inversion identity through q^5 as rational functions of the indeterminate u . It checks the discriminant/product identity through q^9 , computes j through q^5 , and checks inverse modular composition through J^6 .

The first verified coefficient lists are

$$\begin{aligned} a_4(q) &= -5q - 45q^2 - 140q^3 - 365q^4 - 630q^5 + O(q^6), \\ a_6(q) &= -q - 23q^2 - 154q^3 - 647q^4 - 1876q^5 + O(q^6), \\ \Delta(q) &= q - 24q^2 + 252q^3 - 1472q^4 + 4830q^5 + O(q^6). \end{aligned}$$

The formal-parameter sign is checked at $q = 0$:

$$-\frac{X_0(1+h)}{Y_0(1+h)} = \frac{h}{1+h} = h - h^2 + h^3 - h^4 + \dots$$

The test suite also checks finite index-shift instances of the theta functional equation, explicit decreasing exponent witnesses in a rank-two lexicographic group, and sample torsion-degree calculations.

These checks are normalization and transcription guards. They do not prove strong summability, arbitrary-rank surjectivity, an identity at all orders, or a proper-class transport theorem. Those conclusions depend on the mathematical arguments above.

12.2 What an exact symbolic implementation must know

To represent a concrete instance, a symbolic system needs finite expressions for the input Hahn series or a reliable lazy coefficient interface, exact leading exponents and coefficients, and a specified ordered exponent group. The domain test additionally requires deciding whether $|\beta| \leq N\alpha$ for some ordinary N . Comparisons with individual $N\alpha$ are not, by themselves, a terminating decision procedure for that existential statement.

For a lexicographically ordered free finite-rank group, the test is finite. Inspect the first nonzero coordinate of α . Values whose first nonzero coordinate occurs earlier are outside H_α ; values supported at that coordinate or later are inside it. In rational coordinates the finite normalization in theorem 2.2 is then performed by ordinary floor arithmetic at that coordinate, with the lower coordinates determining boundary cases. For a general group, the needed principal-convex-subgroup operation must be part of the representation contract.

For normalized admissible u , an evaluator can return the two isolated rational terms, a lazy positive-tail object, the support certificate (20), and the tail bound (21). That is an exact expression with an explicit precision policy. A finite truncation is not an exact elliptic point unless the discarded tail is retained symbolically. Testing the curve equation numerically after truncation is not a replacement for the exact identity.

The torsion-degree calculation requires membership in $n\Gamma$ and finite cyclic order in $\Gamma/n\Gamma$, which is extra group data. When $\Gamma \cong \mathbb{Z}^r$, (45) supplies a direct algorithm. In a black-box ordered group, it need not be decidable from order comparisons alone.

12.3 A dependency-aware formalization plan

The repository already has a substantial Hahn-support layer, including positive-support evaluation and regrouping [15]. It should be reused. The new proof obligations naturally separate as follows.

Layer	Precise obligations
Ordered groups	Principal convex subgroup, H_{α}^{-} , finite annulus normalization, exact theta-leading-exponent obstruction.
Hahn calculus	Strong coordinate tails, exponent embedding functoriality, coarse coefficient field, unrestricted coarse formal evaluation.
Formal algebra	Tate coefficients, local parameter B and its inverse, projective identity chart, formal-group identities.
Rank-one bridge	Completeness of $K_{H_{\alpha}}$ under its rank-one coarsening and the full classical Tate theorem with explicit hypotheses.
All-rank geometry	Taylor/Hahn agreement, group law in regular charts, coarse reduction diagram, surjectivity and exact kernel.
Arithmetic	Finite coset decomposition of a Hahn extension, character Galois action, primitive torsion-point stabilizer.

This is a plan, not a supplied Lean formalization. In particular, proving the support lemmas while postulating classical Tate uniformization would only establish a theorem conditional on that input. A full formal audit must distinguish that situation from a checked construction of the rank-one theorem. No unchecked code skeleton is included as though it were a machine-verified proof.

13 Limitations and further mathematical questions

The present results settle the internal domain and surjectivity questions posed in section 1. They leave several broader tasks genuinely separate.

Beyond negative valuation of j . The Tate parameter $v(q) > 0$ produces $v(j) < 0$. Curves with nonnegative valuation of j require a different treatment, such as formal groups around a good-reduction model and a residue elliptic curve. The surcomplex field’s algebraic closedness does not by itself provide a single useful transcendental parametrization of that different regime.

Higher-dimensional totally degenerate abelian varieties. A natural extension replaces $q^{\mathbb{Z}}$ by a finite-rank multiplicative period lattice and the quadratic theta exponent by a positive polarization form. One would need an exact ordered-group condition for all lattice directions and a replacement for the single principal convex subgroup H_{α} . The support proof in this paper suggests the right obstruction, but it is not a proof of a general multivariable uniformization theorem.

Intrinsic higher-rank analytic geometry. The algebraic quotient $H_{\alpha}/\mathbb{Z}\alpha$ retains infinitesimal value directions discarded by its Archimedean quotient. Relating it precisely to Hahn analytification, adic spaces, or a suitable skeleton is a separate geometric theorem. The exact sequence (38) is a starting invariant, not a substitute for that theorem.

Functions outside the literal Hahn domain. We proved that the displayed bilateral families fail strong summability outside U_q . We did not prove that every possible resummation, noncanonical assignment, or change of analytic category is impossible. Any proposed extension must specify its summation rule and which identities it preserves; it cannot call the original families strongly Hahn-summable when their leading supports are decreasing.

Independent verification and priority. The most important independent proof checks are the joint-family interchange in theorem 6.3 and the use of regular projective charts in the group law. The source search was not an exhaustive search of valuation theory, formal models, or the model theory of analytic structures. Finding an earlier theorem with these conclusions would change the priority assessment, not the support calculation. The article should be treated as a research draft with explicit proofs, not as an independently certified publication.

14 Conclusion

At arbitrary valuation rank, the correct multiplicative source of the Hahn Tate map is neither automatically all of $K^\times$ nor merely one rank-one coefficient subfield. It is

$$U_q = \{u : v(u) \in H_{v(q)}\}.$$

The restriction is forced by the bilateral support geometry, yet it loses no elliptic-curve points. Classical rank-one uniformization reaches the coarse residue curve; formal evaluation at coarse infinitesimals recovers every remaining point. This proves

$$U_q/q^{\mathbb{Z}} \simeq E_q(K)$$

without claiming convergence at scales where the series do not converge.

The same analysis yields a canonical valuation quotient, exact behavior under scalar extension, support-preserving modular inversion, and an explicit torsion ramification law. In particular, the construction gives a coherent theory of multiplicatively uniformized elliptic curves over $\mathbf{No}[i]$ while retaining its genuinely non-Archimedean scale structure.

A Source and verification audit

A.1 What is attributed, and what is proved here

The principal classical inputs are Tate’s complete rank-one uniformization theorem, the formal Tate curve and its discriminant, Jacobi’s product, algebraic closedness of divisible Hahn fields over $\mathbb{C}$, and standard finite torsion and j -classification of elliptic curves. They are cited rather than presented as discoveries.

The proof supplied here establishes the exact strong-summability domains, the support and precision certificates, unrestricted formal evaluation across a convex coarsening, its agreement with the literal Tate sums, the resulting arbitrary-rank surjective quotient, and the indicated consequences. The finite computations independently check selected identities and signs. They neither replace the classical inputs nor certify the all-rank proof.

A.2 Repository provenance

The reference revision identified during the audit was

a3124af79f66b8b9c196d76b4cbc5ac3938907c4.

The rank-one README was fetched explicitly at that revision. The root README, document map, formalization ledger, and spectral README were read from `main` during the same audit. No claim is made that all manuscripts in the repository were read line by line. In particular, the negative result of a keyword search is treated as a navigation aid, not a formal proof that a topic occurs nowhere.

The delivered files do not alter the repository or claim to be part of its checked Lean build. They form a standalone research package.

A.3 Literature-search scope

The audit checked Tate’s original complete-field formulation and subsequent treatments; the explicit formulas in the 2026 Eterović–Vermeulen paper; rank-one skeleton statements in Baker–Payne–Rabinoff; and the existence and scope of higher-rank Hahn analytification in Foster–Ranganathan. Searches combined Tate uniformization, Hahn fields, higher-rank valuations, and theta series. No matching exact-domain/all-points statement was identified in that limited search. This is not a claim that such a result is absent from all published or unpublished literature.

A.4 Reproduction

The source uses standard TeX Live packages and requires no external figures, bibliography download, or font installation. From the package root, run

```
python code/check_identities.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The first command requires SymPy. The recorded verification output is included so that the finite results remain inspectable without running Python. Recompiling the article does not execute the verification code.

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- [17] *Cofinality, Factorization and Scalar Extension for Entire Hahn Functions at Arbitrary Rank*, research package in [15], docs/surcomplex/entire-functions-at-arbitrary-rank/. The real-valued coarsening at an order unit is its `ent:lem:coarsening`; that report’s standing hypothesis is a set-sized *divisible* exponent group, which the present paper does not assume. This is a repository research draft, not an independently refereed source.